

Course: ENCN423/627 – Sustainable Energy Technologies

Module: Battery Energy Storage Systems (BESS)

Practice problems: Battery structure/processes, Battery choice, applications

Question 1.

The two types of cell ageing (decrease in capacity) are calendar and cycling, give two examples of each

Calendar:

- Chemical decomposition
- Chemical reaction with other cell materials
- Temperature acceleration
- Etc...

Cycling:

- Mechanical stress in electrodes (e.g. heat cycles)
- Formation of solid-electrolyte interface
- Over-charging/under-discharging (overlapping with above)
- Etc...

Question 2.

Which of the following is the key difference between lead-acid and lithium-ion batteries?

- a) Lead-acid batteries must be in constant use
- b) Lead-acid batteries cannot be recharged
- c) In a lead-acid battery, the anode dissolves, while in a lithium-ion battery it does not
- d) In a lithium-ion battery, the anode dissolves, while in a lead-acid battery it does not

Question 3.

What is the difference between a primary and secondary cell?

Primary cells are single-use (non-rechargeable), while secondary cells can be recharged.

Question 4.

One of the applications of BESS in power systems is frequency regulation, describe what this is and what may happen if a power system does not have adequate frequency modulation

Frequency regulation is the maintaining of a power grid's frequency within its stable operating range (in Aotearoa the nominal frequency is 50 Hz). Having a stable frequency in the grid improves stability and equipment efficiency, prevents damage to motors, transformers etc., and reduces operational costs. Remember that if generation > consumption, frequency increases, if consumption > generation, frequency decreases, battery charge/discharge can negate these frequency changes.

Question 5.

What additional component(s) can (typically) be found in

- a) A battery when compared to a cell

A battery houses multiple cells within its casing, and these are connected to form a single unit

- b) A module when compared to a battery

A module houses many batteries in series, and has more detailed methods of control and sensing for charge/discharge regulation

Question 6.

Compare and contrast the strengths and weaknesses of BESS vs. pumped hydro storage technologies

BESS are much quicker to install and more space-efficient than PHES, and are a rapidly advancing technology which tie in easily with increases in renewable generation.

However, PHES have higher energy to power ratios meaning they can store greater energy durations. They also have longer operating lifespans, and can be collocated with hydroelectric generation.

Question 7.

In the context of a power grid, what is energy arbitrage, why is it needed, and how do BESS fill this role in renewable-heavy grids

Energy arbitrage in a power grid is the act of making up for supply and demand mismatches. In a renewable-heavy grid, supply is (mostly) out of the control of the operators, and so tends not to match demand profiles. Therefore, the addition of BESS to such grids can provide the required arbitrage by storing excess energy during peak generation hours, and dispatching it when demand exceeds supply

Question 8.

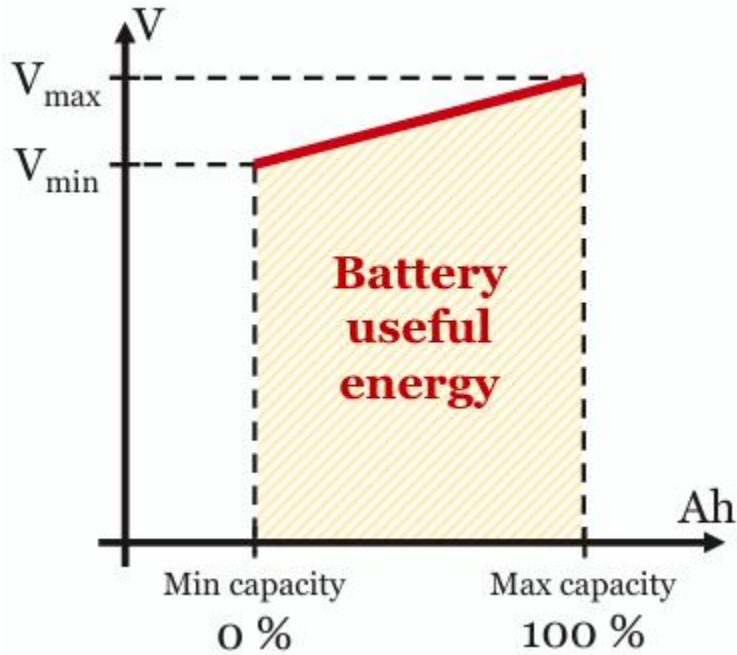
Describe the advantages of sodium-ion batteries when compared to lithium-ion

Sodium-ion batteries are made from a much more abundant material (table salt vs. mined lithium metal), and can therefore also be produced at a much lower cost. Their cold-weather performance is stronger, and they are less prone to thermal runaway (explosions). However, they have a lower energy density and shorter cycle life

Question 9.

Consider a battery with a minimum operational voltage of 250 V, a maximum operational voltage of 315 V, and a capacity of 350 Ah, calculate the amount of useful energy in the battery

Hint: the following plot will be conceptually useful for this and upcoming questions



$$E_{Useful} = \frac{(V_{max} + V_{min})C_{batt}}{2}$$

$$E_{Useful} = \frac{(315 + 250)350}{2} = 98,875 \text{ Wh} = 98.9 \text{ kWh}$$

Question 10.

How would the useful energy of the battery in Question 9 change if a battery controller is applied to maintain an SoC between 10% and 80%?

$$V_{max,use} = V_{max} - \left(\frac{\Delta V (100 - SoC_{max})}{100} \right) \text{ where } \Delta V = V_{max} - V_{min}$$

$$V_{max,use} = 315 - \left(\frac{65(100 - 80)}{100} \right) = 302 \text{ V}$$

$$V_{min,use} = V_{min} + \left(\frac{\Delta V \times SoC_{min}}{100} \right)$$

$$V_{min,use} = 250 + \left(\frac{65 \times 10}{100} \right) = 256.5 \text{ V}$$

$$C_{batt,use} = C_{batt} \left(\frac{\Delta SoC}{100} \right) = 350 \left(\frac{70}{100} \right) = 245 \text{ Ah}$$

$$E_{useful} = \frac{(302 + 256.5)245}{2} = 68,416 \text{ Wh} = 68.4 \text{ kWh}$$

$$\Delta E_{useful} = 30.5 \text{ kWh}$$